Quantum Traffic Gridlock — Technical Report

Code, outputs and run instructions: https://github.com/Harshu3605/quantum-traffic-gridlock-qaoa

1. Traffic model and official cost

Decision bits are x = (A1..A4, B1..B4), with 1 = NS green and 0 = EW green. For each approach k (node n, direction d), the queue follows the PS recursion: $q_{k,t} = q_{k,t-1} + a_{k,t} + u_{k,t} - \min(c, q_{k,t-1} + a_{k,t} + u_{k,t}) \cdot g_{k,t}$. Here g is the green indicator: x for NS and 1−x for EW. The term u is the transfer, and it is non-zero only for B's linked approach: $u_{t+1} = \lfloor f \cdot s^A_t + ½ \rfloor$, where s^A_t is the number of vehicles A's linked approach serves in slot t. The cost is $C(x) = \Sigma_t \Sigma_k [q_{k,t} + w_o \cdot \max(0, q_{k,t} - \text{storage})] + w_s \cdot \#\text{switches}$. Every cost in this report is computed by the organisers' unmodified `official/scorer.py`, which reproduces 116, 146, 237 and 92.5. Our own simulator matches it exactly: on all 256 schedules for 40 random scenarios, including the decimal half-up transfer rounding.

2. QUBO derivation (scenario-independent modelling rule)

Exact decomposition. $C(x) = \Sigma_k D_k(x_{S_k}) + w_s \Sigma_{i,t} (x_{i,t} \oplus x_{i,t-1})$. Each per-approach queue cost D_k depends only on its *support* S_k:

- For three of the four approach groups, S_k is the four bits of the approach's own intersection.
- For B's linked approach, S_k also includes A's four bits, because its inflow is A's linked departures. This dependence carries the A→B coordination.

The decomposition is verified exactly on all 256 schedules (`test_decomposition_is_exact`).

Switch term. This term is already quadratic: $x_t \oplus x_{t-1} = x_t + x_{t-1} - 2x_t x_{t-1}$. It is encoded exactly.

Queue terms. D_k is not quadratic. The min and max operations inside the Lindley recursion create interactions of order up to $|S_k|$. We replace each D_k by its *L2-optimal quadratic surrogate on its own support*. In spin variables $s = 1-2x$, $D_k = \Sigma_{A \subseteq S_k} \hat{c}_A \Pi_{i \in A} s_i$ (Walsh–Fourier expansion), and we keep the terms with $|A| \le 2$. Under the uniform measure, this truncation is the least-squares best quadratic, so no schedule is preferred in the fit. The coefficients $\hat{c}_A = 2^{-|S_k|} \Sigma D_k(s) \cdot \Pi s_i$ are computed by running the single-approach queue recursion over that approach's support cube. Linearity then gives the full QUBO: $Q = \Sigma_k \text{proj}_2(D_k)$ + exact switch terms.

Integrity. The rule is identical for every scenario. It uses only traffic dynamics: arrivals, capacity, storage, transfer and weights. It never ranks schedules, never locates the optimum, and never encodes a target answer, and enumeration of the official cost appears only in the benchmarks. There is one honest caveat. Because the support of B's linked approach is all 8 bits, computing that one approach's projection touches all 256 assignments of that single approach's queue. Summed over approaches, the result equals the global order-2 Walsh truncation of C. It is still a fixed approximation rule that has no knowledge of the minimizer.

Size. 8 binary variables, 8 qubits, no auxiliary variables. The fully linked approach makes the coupling graph complete, with 28 ZZ terms. The Ising form uses $x = (1-z)/2$.

Penalties. Overflow is soft, as specified, with weight w_o. A multiplier λ_o on the overflow part is kept as an ablation knob, default 1, so the QUBO matches the official weights.

3. QAOA design

- **Ansatz.** $|+\rangle^8$, then p layers of $e^{-i\gamma H_C}$ (RZZ and RZ gates) followed by an RX(2β) mixer. Default p = 4, the maximum allowed. Training normalises the cost to zero mean and unit standard deviation over the cube, so the γ ranges are scenario-independent. The physical angles γ/σ are bound into the exported circuit.
- **Training.** Exact numpy statevector for 256 amplitudes. The objective is the expected QUBO energy, because lift is a mean-cost metric.
- **Initialisation (INTERP).** Score 7 linear-ramp (TQA) starts at p = 1. Then for each depth d = 1…p, run COBYLA and linearly interpolate the angles to depth d+1. All of this shares a hard **300-evaluation** budget that the code enforces. The seed adds a small jitter to the starting angles.
- **Sampling and consistency.** Final sampling uses Qiskit `StatevectorSampler(seed)` with exactly 4,096 shots. We check that the Qiskit statevector of the bound circuit matches the trained numpy distribution (TVD ≈ 1e−15). A test also re-imports `circuit.qasm` and checks it against `counts.json`.
- **Resources.** One run takes about 0.4–2 s on CPU, far below the 10-minute limit.

4. Results (5 seeds: 0–4; costs are official costs)

	R	H_sample (synthetic)
(1) best alternating	116.0 (1010|1010)	194.0

(2) greedy (larger queue, ties NS)	92.5 (0111|1001)	188.0
(3) exact optimum C*	**72.0** (1110|1010)	**183.0** (1001|0011)
(4) uniform-random mean C_rand	151.25	247.65
(5) QUBO exact minimizer (surrogate ceiling)	72.0, regret 0 %	188.0, regret 2.7 %
corr(QUBO, official) over 256 schedules	0.995	0.991
(6) raw QAOA E_Q (mean over seeds)	88.2	194.5
lift: mean / **worst seed**	0.796 / **0.792**	0.822 / **0.819**
elite enrichment (13 best)	12.55×	10.61×
P(optimal schedule) per shot	8.3 %	8.5 %
hybrid QAOA + bit-flip local search (labelled, not raw)	72.0	183.0

On both scenarios, raw QAOA's mean sample cost beats the alternating baseline. Its mean does not beat greedy, which is a single deterministic schedule: on R, E_Q = 88.2 against greedy's 92.5, but on H_sample E_Q is 194.5, worse than greedy's 188. About 8 % of shots hit the exact optimum.

Generalisation. 30 random scenarios from the published hidden ranges, with a fixed generator seed of 2026; files are in `outputs/experiments/random_scenarios`.

- QUBO regret: mean 0.9 %, at most 5 % on 93 % of scenarios, and exactly zero on 60 %.
- Worst case: 10.6 % (rand17). The next worst is 8.0 %.
- Lift: mean 0.84. The worst seed's lift has a median of 0.85 and a minimum of 0.64.
- Enrichment: mean 12.3×.
- Greedy's mean regret is 5.4 %.
- The labelled hybrid reached C* on all 30 scenarios for every seed.

Ablations (5 seeds each; worst-seed lift shown).

p	1	2	3	4
R, INTERP	0.613	0.729	0.772	**0.792**
R, linear ramp only	0.613	0.687	0.692	0.515
R, random init	0.613	0.593	0.591	0.597
H_sample, INTERP	0.504	0.704	0.784	**0.819**
H_sample, random init	0.504	0.498	0.513	0.509

Only INTERP improves steadily with depth. Without layer-wise warm starts, COBYLA with 300 evaluations gets stuck at $p \geq 3$. The overflow multiplier $\lambda_o \in \{0.5, 1, 1.5, 2\}$ changes lift by less than 0.02. On H_sample, $\lambda_o = 0.5$ happens to remove the 2.7 % regret. We kept $\lambda_o = 1$ because tuning it on the evaluation scenario would be overfitting.

5. Limitations (honest)

- The quadratic surrogate is approximate. Regret can reach about 10 % when queue saturation creates strong higher-order interactions. A fix with auxiliary slack variables (≤ 8 spare qubits) is a possible extension that we did not implement.
- Training uses the exact noiseless statevector, which is legitimate in simulation because training shots are optional. We performed no noise or hardware study.
- At 8 qubits, enumeration is trivial, so we claim no quantum advantage. QAOA here is a sampler whose distribution is strongly concentrated on low-cost schedules: about 12× elite enrichment.
- H_sample is synthetic. The official H (released at hour 18) and S must be run with the unchanged code.